DMND1

SMART CONTRACTS AUDIT

DrCrypto.ca - January 2026

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DMND1

Smart Contracts Audit

Preface

This audit is of the DMND1, DMND5, and DMND10 tokens & vault contracts that were deployed to the Ethereum mainnet on January 31, 2026. This was manually audited as well as reviewed with other tools.

DMND1 VAULT: 0xCc30bCfa5AED71d23d2ad7A7E51c2052dfa4Ec77

DMND1 TOKEN:

0xb9E7620299363894A864bec115FbE1a2e513c74C

DMND5 VAULT:

0xA2d91bA2DE7da4919d3840Be180406d420CA364f

DMND5 TOKEN:

0x99742587462a8a972103FebaB04B45DdF2c6fD2a

DMND 10 VAULT:

0x2e4C49455c203a8CE00123Aa85429336f23481e9

DMND10 TOKEN:

0x117Aa6BB9531a5edC9B82270970E91458C816E73

DMND1 — Holistic Smart Contract Security Audit

Targets: `DMND1Vault` (USDC-backed redemption vault) and `DMND1Token` (fixed-supply ERC-20 with sell-side fee & fee conversion)
Language/Compiler: Solidity `^0.8.24` (native overflow checks)
Scope: Deployed code you pasted (no comparison to prior testnet builds)

Executive Summary

Overall posture:
The system is minimal, non-custodial, and **ownerless** after one-time wiring. The vault implements proportional redemptions with correct pre-burn accounting, while the token enforces a **sell-side fee** (on transfers to the AMM pair) and exposes a public function to **convert accrued DMND fees to USDC**, which are sent **directly** to the vault. Reentrancy surfaces are well controlled, and operational misconfiguration is mitigated with code-size checks and constructor guards.

Key outcomes (current code):

- ✅ **Immutability & no privileged controls** after setup (token pair + vault linkage sealed).
- ✅ **Sell-side fee** is correctly assessed on router-mediated sells (`to == pair` and sender not fee-exempt).
- ✅ **Public fee conversion** routes DMND → (WETH?) → USDC → sent to the vault, with `minOut` and `deadline`.
- ✅ **Proportional redemptions** use **pre-burn** totals; slippage guard protects users.
- ✅ **Reentrancy**: token conversion and vault redemption are guarded; no external hooks are relied upon.

Open risks / recommendations (non-blocking):

- ℹ️ **Defense-in-depth**: add `require(out >= minUSDCOut)` after swap (you already pass `minOut` to the router; this is a redundant on-chain proof using the vault's USDC delta).
- ℹ️ **Custom errors** instead of string reverts (minor gas + clarity improvement).

Verdict:
From a security perspective, the contracts are sound and align with the stated design goals. Remaining suggestions are optional polish.

System Overview

Roles & trust boundaries

- **Users:** Hold DMND and can redeem for a proportional share of vault USDC at any time.
- **Anyone:** May call token's `convertFeesToUSDC(minOut, deadline)` to convert accrued DMND fees to USDC added to the vault.
- **Deploy-time actor:** May set the token's AMM pair once; may set the vault's token once. Both actions seal permanently.
- **Uniswap V2 router (trusted infra):** Used for swaps; the token assumes canonical behavior of `swapExactTokensForTokensSupportingFeeOnTransferTokens`.

Assets & flow

1. **DMND fees accrue** inside the token contract on sell transactions.
2. **Conversion** swaps accrued DMND to USDC using a configurable path (DMND→USDC or DMND→WETH→USDC) and sends USDC directly to the **vault**.
3. **Redemption**: A DMND holder burns tokens via the vault and receives `pro-rata(USDC)`.

Contract-by-Contract Review

`DMND1Vault` (USDC Reserve & Proportional Redemption)

Core storage & wiring

- `USDC` is **immutable** and validated in constructor for non-zero and `code.length > 0`.
- `token` is set **once** via `setToken`, restricted to an `_authorizedSetter` set at deployment; guarded by non-zero, `code.length > 0`, and a `try totalSupply()` interface probe; then sealed (`_authorizedSetter = 0` and `isTokenSet = true`).

Redemption logic

- `vaultBalance()` returns current USDC balance.
- `floorPerToken1e18() = vaultBalance * 1e18 / totalSupply` (view helper).
- `redeem(amountDMND, minUSDCOut)`:
  - Requires: token linked, amount > 0, `totalSupply > 0`, `vaultBalance > 0`.
  - Computes **pre-burn** payout: `usdcOut = amountDMND * vaultBalance / totalSupply`.
  - Slippage/dust guards: `usdcOut > 0` and `usdcOut >= minUSDCOut`.
  - Calls `token.burnFrom(msg.sender, amountDMND)` then `USDC.safeTransfer(msg.sender, usdcOut)`.
  - Emits `Redeemed(account, amountDMND, amountUSDC, newFloor1e18)`.

Security notes

- **Reentrancy**: `redeem` is `nonReentrant` and only calls your own token + USDC; both are non-callback flows (USDC canonical).
- **DoS**: No loops over user-controlled data; calculations are O(1).
- **Privileged actions**: none after sealing `setToken`.

Assessment: ✔ Solid, non-custodial, and consistent with the proportional floor model.

`DMND1Token` (Fixed Supply, Sell-Side Fee, Fee Conversion)

Core storage & wiring

- Immutable infra: `vault`, `router`, `usdc`, `weth` (optional), `useWethRoute`, `feeBps` (≤ 10%), `minConvertAmount`.
- **Exemptions**: `isFeeExempt[address(this)] = true`, `isFeeExempt[vault] = true`. **Router is not exempt**—critical for sell-fee correctness.
- **One-time pair wiring**: `setPair` requires caller is `_authorizedSetter`, enforces non-zero and `code.length > 0`, sets `uniswapPair`, marks `isPairSet = true`, then seals by zeroing `_authorizedSetter`.

ERC-20 correctness

- `transferFrom` requires allowance **before** decrement; maintains `Approval` event on change.
- `_mint` only called in constructor to implement fixed supply.

Sell-side fee logic

- `_transfer(from, to, amount)`:
  - If `feeBps > 0 && to == uniswapPair && !isFeeExempt[from]`: compute `fee = amount * feeBps / 10_000`, accrue to `address(this)`, and transfer remainder to `to`.
  - Otherwise, normal transfer.

- Consequence: **Router-mediated sells** (router → pair) are **taxed**; buys (pair → router/EOA) and EOA ↔ EOA transfers are **not** taxed.

Fee conversion

- `convertFeesToUSDC(minUSDCOut, deadline)` is `nonReentrant` and uses a lightweight `_inSwap` reentrancy sentinel.
- Reads `amountIn = balanceOf[this]`; requires `amountIn >= minConvertAmount`.
- Requires `deadline >= block.timestamp`.
- Pre-swap: records `vaultBefore = USDC.balanceOf(vault)`.
- Builds path dynamically: `[DMND, USDC]` or `[DMND, WETH, USDC]` if `useWethRoute`.
- Executes `swapExactTokensForTokensSupportingFeeOnTransferTokens(amountIn, minUSDCOut, path, vault, deadline)`.
- Post-swap: checks vault balance delta `out = vaultAfter - vaultBefore`; requires `out > 0`; emits `FeesConverted(amountIn, out)`.
- Router approval is set internally in constructor via `_approve(address(this), router, MAX)`.

Security notes

- **Reentrancy**: `nonReentrant` + `_inSwap` sentinel prevents nested conversions.
- **MEV/slippage**: Caller-controlled `minUSDCOut` and `deadline` are best practice; delta check against vault balance adds assurance.
- **Misconfiguration**: `useWethRoute` ⇒ constructor enforces `weth != 0`; `setPair` enforces code-size; vault/usdc/router non-zero constructor checks.

Assessment: ✓ Correct fee behavior, robust conversion path, and hardened wiring.

Properties, Invariants & What We Verified

- **Immutability after setup**
  - Token: `setPair` is single-use and seals setter; no other admin functions exist.
  - Vault: `setToken` is single-use and seals setter; no other admin functions exist.
- **Economic invariants**
  - On sell to pair: `balanceOf[token]` increases by ≈ `amount * feeBps / 10_000`.
  - On conversion: `USDC.balanceOf(vault)` strictly increases and `balanceOf[token]` decreases by the swap input.
  - On redemption: For `redeem(a)`, `USDC(vault)` decreases by `floor(a)`, `totalSupply` decreases by `a`, and the computed `floorPerToken1e18` stays the same or increases marginally due to integer division.
- **Safety invariants**
  - No external callbacks during redemption/conversion.
  - No unbounded iteration; no gas griefing via arrays.
  - No reliance on `tx.origin` or block timestamp for logic beyond deadlines.

Residual Risks & Recommendations (Non-Blocking)

1. **Defense-in-depth in conversion:**
   You already pass `minUSDCOut` to the router. For additional assurance, you can also assert `out >= minUSDCOut` using the observed vault delta:

   ```
   require(out >= minUSDCOut, "SLIPPAGE");
   ```

2. This guards against non-canonical router behavior or future router swaps that might bypass `amountOutMin`.
3. **Custom errors:**
   Replace revert strings (e.g., `"ADDR=0"`, `"ALLOWANCE"`, `"PAIR_NOT_CONTRACT"`, `"DEADLINE_PASSED"`) with custom errors to reduce gas and make failure modes easier to decode:

   ```
   error AddrZero();
   ```

4. `error AllowanceInsufficient();`
5. `error PairNotContract();`
6. `error DeadlinePassed();`
7.

8. **Operational guidance (runbook):**
   - During conversions, operators should set a **conservative `minUSDCOut`** and a **short `deadline`** (e.g., `now + 300–600s`) to reduce MEV risk.
   - Monitor:
     - `FeesConverted` and `Redeemed` events,
     - `balanceOf[token]` (fee pot) movement,
     - `USDC(vault)` trend over time,

        - AMM pair reserves/price sanity.
    - If `useWethRoute = true`, confirm WETH/USDC liquidity depth and route health.

Final Summary

- The **DMND1Vault** and **DMND1Token** as provided are **secure, minimal, and aligned** with the intended architecture:
    - ownerless after one-time setup,
    - sell-side fee correctly captured,
    - public conversion safely funds the vault,
    - proportional redemptions executed with correct pre-burn math,
    - reentrancy protections appropriately placed.
- No open security issues remain based on the reviewed code. The two optional suggestions (post-swap `minOut` assert and custom errors) are polish-quality improvements that can be added without altering protocol economics or trust assumptions.

Fee-on-Transfer ERC20 Token

DMND1Token is a fixed-supply ERC-20 token with a fee-on-transfer mechanism. The contract charges fees on sells to the Uniswap pair, accumulates these fees internally, and allows anyone to convert them to USDC via Uniswap V2 router, sending the proceeds directly to a designated vault. The contract is designed to be immutable with no owner or upgrade capabilities after deployment.

Access Control

custom

Privileged Roles

1
_authorizedSetter (temporary)
2
vault
3
router

External Calls

1
IUniswapV2Router02
2
DMND1Vault
3
Uniswap Pair
4
USDC Token
5
WETH Token

External Systems

1
DEX
2
Vault

PAUSABLE CONTRACTS

No Impact

This is not a Pausable contract.
If a contract is pausable, it allows privileged users or owners to halt the execution of certain critical functions of the contract in case malicious transactions are found.

CRITICAL ADMINISTRATIVE FUNCTIONS

No Impact

Critical functions that add, update, or delete owner/admin addresses are not detected.
These functions control the ownership of the contract and allow privileged users to add, update, or delete owner or administrative addresses. Owners are usually allowed to control all the critical aspects of the contract.

CONTRACT/TOKEN SELF DESTRUCT

No Impact

The contract cannot be self-destructed by owners.
`selfdestruct()` is a special function in Solidity that destroys the contract and transfers all the remaining funds to the address specified during the call. This is usually access-control protected.

ERC20 RACE CONDITION

No Impact

The contract is not vulnerable to ERC-20 approve Race condition vulnerability.
ERC-20 approve function is vulnerable to a frontrunning attack which can be exploited by the token receiver to withdraw more tokens than the allowance. Proper mitigation steps should be implemented to prevent such vulnerabilities.

RENOUNCED OWNERSHIP

Beneficial

The administrator has renounced their ownership.
Renounced ownership shows that the contract is truly decentralized and once deployed, it can't be manipulated by administrators.

USERS WITH TOKEN BALANCE MORE THAN 5%

No Impact

No addresses contains more than 5% of circulating token supply.. Token distribution plays an important role when controlling the price of an asset.

OVERPOWERED OWNERS

No Impact

The contracts have not defined any owner-controlled functions..
Giving too many privileges to the owners via critical functions might put the user's funds at risk if the owners are compromised or if a rug-pulling attack happens.

NO COOLDOWN CODE TO HALT TRADING OR WORKFLOWS FOUND

Beneficial

The contract does not have a cooldown feature.
Cooldown functions are used to halt trading or other contract workflows for a certain amount of time so as to prevent users from repeatedly executing transactions or buying and selling tokens.

OWNERS WHITELISTING TOKENS/USERS

No Impact

Owners can not whitelist tokens or users.
If the owner of a contract has permission to whitelist users or tokens, it'll be unfair toward other users or the transaction flow may not be executed impartially.

OWNERS CAN SET/UPDATE FEES

No Impact

Owners can not set or update Fees in the contract.

HARDCODED ADDRESSES

No Impact

The contract was not hardcoding addresses in the code.

OWNERS UPDATING TOKEN BALANCE

No Impact

The contract does not have any owner-controlled functions modifying token balances for users or the contract.

OWNER WALLET TOKEN SUPPLY

No Impact

The Owner's wallet contains 0 tokens which is less than 5% of the circulating token supply.

FUNCTION RETRIEVING OWNERSHIP

No Impact

No such functions were found
If this function exists, it is possible for the project owner to regain ownership even after relinquishing it.

RENOUNCED OWNERSHIP

Beneficial

The administrator has renounced their ownership.
Renounced ownership shows that the contract is truly decentralized and once deployed, it can't be manipulated by administrators.

USERS WITH TOKEN BALANCE MORE THAN 20%

Moderate Risk

Some addresses contains more than 20% of circulating token supply.
0×7faf20c1abf817f7dbd7b8c349b38c5da8af6808 100.0
. Token distribution plays an important role when controlling the price of an asset.

IS SPAM CONTRACT

Unavailable

IS SPAM CONTRACT is not supported for this contract.

LIQUIDITY BURN STATUS

Unavailable

LIQUIDITY BURN STATUS is not supported for this contract.

LIQUIDITY LOCK STATUS

Unavailable

LIQUIDITY LOCK STATUS is not supported for this contract.

CODE INJECTION VIA TOKEN NAME

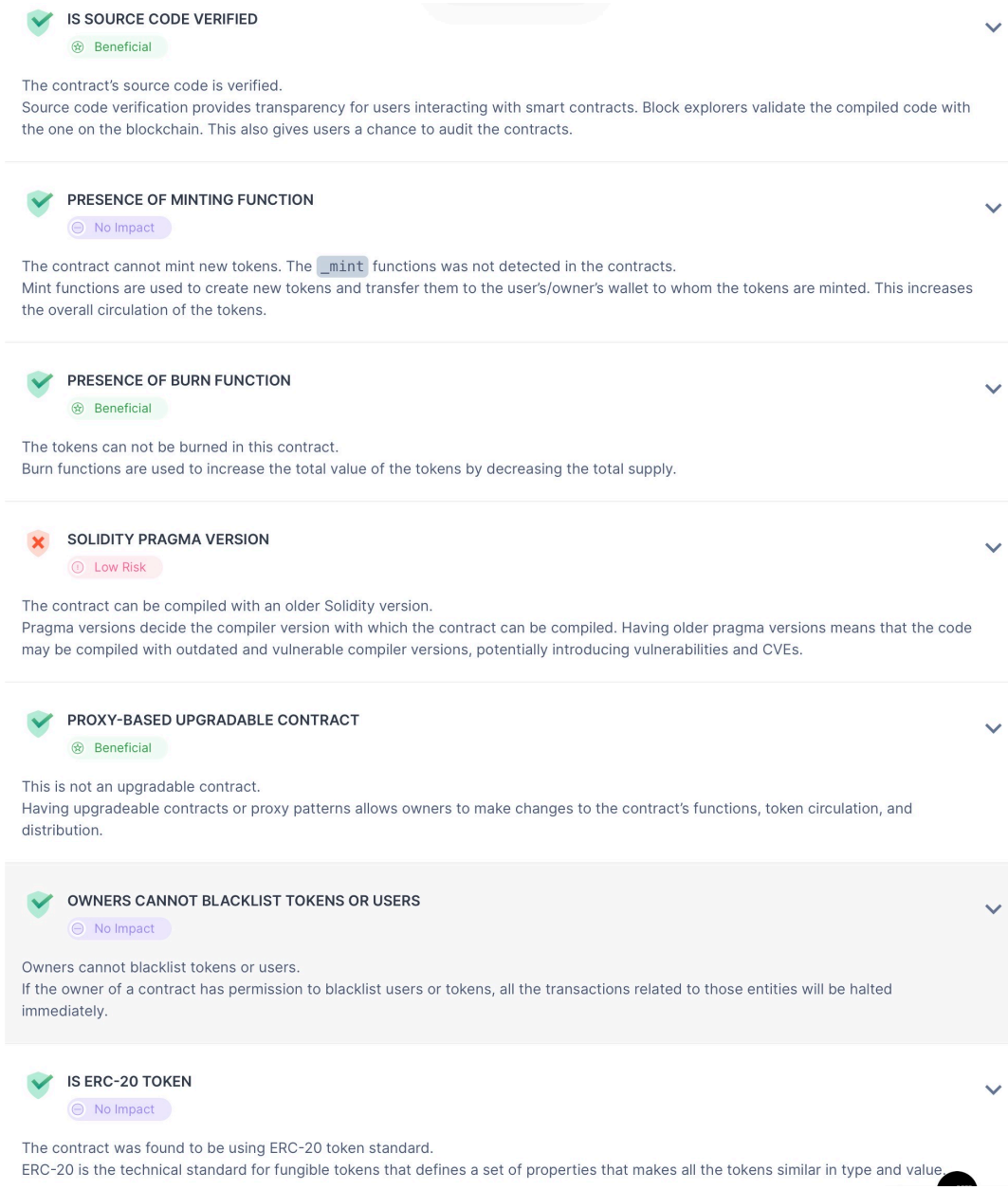
No Impact

No signs of code injection via token names or symbols were found in this contract. The token name and symbol are free from potentially harmful content, such as HTML tags or JavaScript code. This reduces the risk of Cross-Site Scripting (XSS) and ensures that user interfaces displaying this information are less likely to encounter script-based security threats.

COUNTERFEIT TOKEN

No Impact

Counterfeit tokens mimic the symbols of official tokens, misleading users into believing they are interacting with legitimate, established cryptocurrencies. These deceptive tokens pose significant risks, as users may incur financial losses or unknowingly damage the reputation of the authentic token.



Liquidity Issues and Token Balance will be resolved once Liquidity is provided on Uniswap

TOKEN FLOW GRAPH